



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Software Requirements Specifications

Hospital Management System (HMS)

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1.0 Introduction

1.1 Purpose

The purpose of this document is to specify the functionality, interface, data, performance, security, and quality requirements of the Hospital Management System (HMS). The specification is derived from the existing application source code and database schema.

This document is intended for system developers, testers, project managers, maintainers, and client representatives who need a shared understanding of the HMS behavior.

1.2 Scope

HMS is a web-based hospital appointment and administration system implemented using PHP, MySQL, HTML, CSS, JavaScript, Bootstrap, and TCPDF. The system allows patients to register and book appointments, doctors to view appointments and issue prescriptions, and administrators to manage doctors and view operational records.

HMS shall support the following high-level operations:

1. Patient registration and patient login.
2. Doctor login.
3. Administrator login.
4. Appointment booking with doctor specialization, doctor selection, fee display, date selection, and time selection.
5. Appointment cancellation by patient and doctor.
6. Prescription creation by doctor.
7. Prescription viewing by patient, doctor, and administrator.
8. Bill PDF generation for patient prescriptions and appointment fees.
9. Administrator viewing and searching of doctors, patients, appointments, prescriptions, and contact messages.
10. Administrator adding and deleting doctor accounts.
11. Public contact message submission.
12. Public informational pages for home, about/services, and contact.

1.3 Definitions and Acronyms

Term	Definition
Admin	A privileged staff user who can view system records and manage doctors. The source code uses both "Admin" and "Receptionist" labels for this role.
Appointment	A scheduled meeting between a patient and a doctor on a selected date and time.
Doctor	A clinical user who can view assigned appointments, cancel appointments, and create prescriptions.
HMS	Hospital Management System.
Patient	A user who registers, logs in, books appointments, views appointments, views prescriptions, and generates bills.
Prescription	A clinical record created by a doctor containing disease, allergy, and prescription information for a patient appointment.
TCPDF	PHP library used by HMS to generate PDF bills.
UI	User Interface.

Table 1.0 Definitions and Acronyms

1.4 References

1. <https://github.com/kishan0725/Hospital-Management-System.git>
2. <https://www.scribd.com/document/529458410/01-TAB-2-ATMS-Software-Requirements-Specifications-v1-0-0>

1.5 Overview

Section 1 introduces the purpose, scope, definitions, and references for HMS.

Section 2 describes the overall product perspective, major features, users, environment, constraints, assumptions, and dependencies.

Section 3 defines the specific external interface, functionality, data, performance, and software quality requirements.

2.0 Overall Description

2.1 Product Perspective

HMS is a web-based hospital management system that has been developed in PHP for server-side programming, MySQL/MariaDB for database storage, and HTML, CSS, JavaScript and Bootstraps for the user interface. The system provides an interface to the user's web browser and runs its core logic on a web server to handle the application logic, create and read from the database, and generate role-based pages and functions. The system also uses TCPDF to produce bills in the PDF format. In real life, the project will use a XAMPP Web server with Apache and MySQL services.

2.1.1 User Interfaces

Table 2.0 shows the summarizes of the graphical user interface requirements specified for HMS:

Requirement ID	Description	Priority
REQ_UI001	HMS shall provide a home page with patient registration, doctor login, and admin login interfaces.	High
REQ_UI002	HMS shall provide a separate patient login page for already registered patients.	High
REQ_UI003	HMS shall provide a patient dashboard with functions such as booking appointments, viewing appointment history, viewing prescriptions, and bill-related actions.	High

REQ_UI004	HMS shall provide a doctor dashboard with functions such as for viewing appointments, searching appointments, and managing prescriptions.	High
REQ_UI005	HMS shall provide an admin dashboard with functions such as viewing records, managing doctors, and handling user queries.	High
REQ_UI006	HMS shall include confirmation and error messages for important activities like unsuccessful login, appointment booking and cancellation, doctor profile creation and deletion, and messaging contact form.	Medium
REQ_UI007	HMS shall use Bootstrap-based responsive layouts for forms, tables, navigation bars, buttons, and dashboards.	Medium

Table 2.0 HMS GUI Requirements

2.1.2 Hardware Interfaces

HMS does not have any direct interface with medical devices in this current version. The device used in the system is a typical computer that supports any modern web browser (e.g. laptop, desktop PC). Users interact with the system through standard input devices (keyboard, mouse) and output (monitor). In the server side, a computer running Apache, PHP and MySQL/MariaDB is needed.

2.1.3 Communication Interfaces

HMS shall have communication between client (browser) and server (web server) in accordance with the normal request-response pattern of HTTP. Users interact with the web server through web forms like registering, logins, appointments, contacts and searching, and the server responds with the appropriate processing and extraction or modification of data stored in the MySQL/MariaDB database. This communication is usually via XAMPP in a local deployment environment via Apache and localhost.

2.2 Product Features

Table 3.0 shows the list of features to be implemented in HMS with separate roles.

Feature ID	Feature	Description	Accessible Role
F001	Register Account	Allows a new patient to register by entering personal details such as first name, last name, email, etc.	Patient
F002	Login	Allows a patient, doctor, or admin to log in using valid credentials and access the corresponding dashboard.	Patient, Doctor, Admin

F003	Logout	Allows logged-in user to terminate the current session and return to the login page.	Patient, Doctor, Admin
F004	Book Appointment	Allows patients to select specialization, doctor, fee, future date, and time slot to create an appointment.	Patient
F005	View Appointment History	Allows patients to view appointment history, including doctor name, consultation fee, date, time, and appointment status.	Patient
F006	Cancel Appointment by Patient	Allows patients to cancel an active appointment.	Patient
F007	View Prescription	Allows patients to view prescription details, perform bill payment, and generate a bill in PDF format.	Patient
F008	Submit Contact Query	Allows users to submit name, email, contact	Patient

		number, and message through the contact page.	
F009	View Doctor Appointments	Allows doctors to view appointments assigned to them.	Doctor
F010	Prescribe Medication	Allows doctors to enter disease, allergy, and prescription information for a selected appointment.	Doctor
F011	Cancel Appointment	Allows doctors to cancel an active appointment assigned to them.	Doctor
F012	View Prescriptions List	Allows doctors to view prescription records associated with their patients.	Doctor
F013	Add Doctor	Allows admin to add a new doctor account into the system.	Admin
F014	Delete Doctor	Allows admin to delete an existing doctor account from the system.	Admin
F015	View & Search Doctor List	Allows admins to view all registered	Admin

		doctors and search for a doctor by email.	
F016	View & Search Patient List	Allows admin to view all registered patients and search for a patient by contact number.	Admin
F017	View & Search Appointment Details	Allows admins to view all appointments and search appointment records by patient contact number.	Admin
F018	View Prescription List	Allows admins to view all prescription records in the system.	Admin
F019	View & Search User Queries	Allows admins to view submitted queries and search them by contact number.	Admin

Table 3.0 HMS Features List

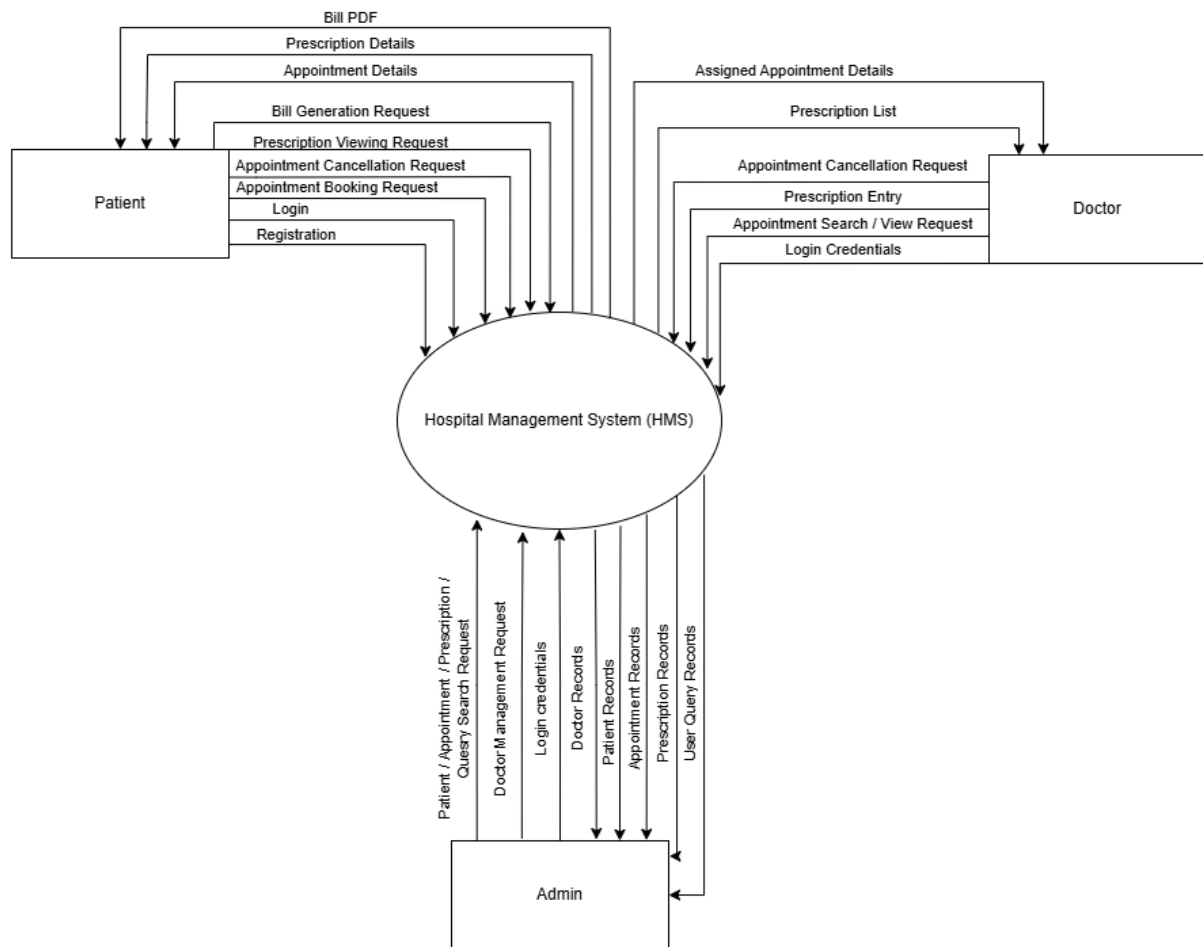


Figure 1.0 HMS Context Diagram

Figure 1.0 shows the context diagram of the Hospital Management System (HMS). It communicates with three external actors, the Patient, Doctor, and Admin. Users can register and login, schedule and cancel appointments, view prescriptions and bills. Doctors log into the system to view the appointments assigned to them, view and manage prescription details, and cancel appointments where needed. Admins use the system to access the system, manage medical doctor accounts, and view or search the information about the following operations: patient, appointment, prescription, and query.

2.3 User Characteristics

HMS primarily focuses on the patient, doctor, and admin user roles. Table 4.0 shows that they use the systems in different ways for operating the systems and hence have different knowledge and access requirements.

User Role	Description	Required Knowledge
Patient	A registered user who logs into the HMS, books appointments, views appointment history, prescription and prints bills.	Basic computer and web application knowledge, including login, form submission, and menu navigation.
Doctor	A logged on medical staff user who uses HMS to view appointments assigned to them, to search for appointments, to cancel appointments when needed and to write prescriptions.	Basic computer skills and familiarity with appointment handling and prescription entry.
Admin	A logged-on system administrator who accesses the HMS system to update doctor details and view business data including patient lists, doctor lists, appointment lists, prescriptions and contact requests.	Moderate computer skills and basic understanding of hospital administrative workflow and record management.

Table 4.0 Intended Users & Required Knowledge

3.0 Specific Requirements

3.1 External Interface

In this section, we explain the external interface requirements of the HMS, grouped by the main HMS interfaces.

3.1.1 Home and Registration Interface

Requirement ID	REQ_IO101	Version	1.0
Item	Home Navigation Interface (Output)		
Description	Main navigation interface displayed on the HMS home page.		
Purpose	To allow users to access the main website sections and available login or registration areas.		
Format	Web page navigation bar	Valid Range	Home, About/Services, Contact
Related I/O	REQ_IO102, REQ_IO103, REQ_IO104		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO102	Version	1.0
Item	Patient Registration Form (Input)		
Description	Registration form used by a new patient to create an account in HMS.		
Purpose	To allow a patient to register a new account.		
Format	Web form	Valid Range	First name, last name, gender, email, contact number, password, confirm password
Related I/O	REQ_IO103		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO103	Version	1.0
Item	Patient Registration Credentials (Input)		
Description	Credentials entered by the patient during registration, including email, password, and confirm password.		
Purpose	To create authentication details for future patient login.		
Format	Email field and password fields	Valid Range	Valid email format; password length of at least 6 characters; confirm password must match password
Related I/O	REQ_IO102		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO104	Version	1.0
Item	Existing Patient Login Link (Input)		
Description	Link on the home page that redirects an already registered patient to the patient login page.		
Purpose	To provide access to the patient login interface.		
Format	Hyperlink / Button	Valid Range	Not applicable
Related I/O	REQ_IO201		
Author	PANG TIAN YOU		

3.1.2 Patient Login Interface

Requirement ID	REQ_IO201	Version	1.0
Item	Patient Login Form (Input)		
Description	Login form used by a registered patient to access the HMS patient dashboard.		
Purpose	To authenticate a patient and allow access to patient functions.		
Format	Web form	Valid Range	Registered email address and matching password
Related I/O	REQ_IO202		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO202	Version	1.0
Item	Patient Login Credentials (Input)		
Description	Patient email and password entered during login.		
Purpose	To verify patient identity before access is granted.		
Format	Email field and password field	Valid Range	Valid registered patient email and correct password
Related I/O	REQ_IO201		
Author	CHIN ZHEN HO		

3.1.3 Doctor Login Interface

Requirement ID	REQ_IO301	Version	1.0
Item	Doctor Login Form (Input)		
Description	Login form used by a doctor to access the doctor's dashboard.		
Purpose	To authenticate a doctor and allow access to doctor-specific functions.		
Format	Web Form	Valid Range	Registered doctor username and matching password
Related I/O	REQ_IO302		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO302	Version	1.0
Item	Doctor Login Credentials (Input)		
Description	Doctor username and password entered during login.		
Purpose	To verify doctor identity before system access is granted.		
Format	Text field and password field	Valid Range	Existing doctor username and correct password
Related I/O	REQ_IO301		
Author	LIM KAI SHEN		

3.1.4 Admin Login Interface

Requirement ID	REQ_IO401	Version	1.0
Item	Admin Login Form (Input)		
Description	Login form used by an admin to access the HMS admin dashboard.		
Purpose	To authenticate an admin and allow access to administrative functions.		
Format	Web Form	Valid Range	Registered admin username and matching password
Related I/O	REQ_IO402		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO402	Version	1.0
Item	Admin Login Credentials (Input)		
Description	Admin username and password entered during login.		
Purpose	To verify admin identity before access to the admin dashboard is granted.		
Format	Text field and password field	Valid Range	Existing admin username and correct password
Related I/O	REQ_IO401		
Author	LIM KAI SHEN		

3.1.5 Appointment Booking Interface

Requirement ID	REQ_IO501	Version	1.0
Item	Doctor Specialization (Input)		
Description	Specialization selected by the patient when booking an appointment.		
Purpose	To filter and identify the relevant doctors for appointment booking.		
Format	Drop-down list	Valid Range	Existing specialization values stored in the system
Related I/O	REQ_IO502		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO502	Version	1.0
Item	Doctor Selection (Input)		
Description	Doctor chosen by the patient for the appointment.		
Purpose	To assign a specific doctor to the appointment record.		
Format	Drop-down list	Valid Range	Existing doctor records in the system
Related I/O	REQ_IO501, REQ_IO503		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO503	Version	1.0
Item	Consultation Fee Display (Output)		
Description	Consultation fee displayed based on the doctor selected by the patient.		
Purpose	To inform the patient of the doctor's fee before appointment confirmation.		
Format	Numeric display field	Valid Range	Existing doctor fee value

Related I/O	REQ_IO502
Author	PANG TIAN YOU

Requirement ID	REQ_IO504	Version	1.0
Item	Appointment Date (Input)		
Description	Date selected by the patient for the appointment.		
Purpose	To specify the appointment date.		
Format	Date field	Valid Range	Future date only
Related I/O	REQ_IO505		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO505	Version	1.0
Item	Appointment Time (Input)		
Description	Time slot selected by the patient for the appointment.		
Purpose	To specify the appointment time.		
Format	Drop-down list	Valid Range	08:00, 10:00, 12:00, 14:00, 16:00
Related I/O	REQ_IO504, REQ_IO506		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO506	Version	1.0
Item	Book Appointment Action (Input)		
Description	Submission action used to create a new appointment entry after all required booking data has been selected.		
Purpose	To send booking information to the system for validation and creation of an appointment record.		
Format	Button	Valid Range	Not applicable
Related I/O	REQ_IO501, REQ_IO502, REQ_IO503, REQ_IO504, REQ_IO505		
Author	PANG TIAN YOU		

3.1.6 Patient Appointment History Interface

Requirement ID	REQ_IO601	Version	1.0
Item	Appointment History List (Output)		
Description	List of appointment records displayed on the patient dashboard.		
Purpose	To allow a patient to view previously created appointments and their current status.		
Format	Table	Valid Range	Doctor name, consultation fee, appointment date, appointment time, status, action
Related I/O	REQ_IO602		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO602	Version	1.0
Item	Appointment Status Display (Output)		
Description	Current status shown for each appointment record.		
Purpose	To inform the patient whether an appointment is active or cancelled.		
Format	Text / status field	Valid Range	Active, Cancelled
Related I/O	REQ_IO601, REQ_IO603		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO603	Version	1.0
Item	Cancel Appointment Action (Input)		
Description	Action available to the patient for cancelling an active appointment.		
Purpose	To allow cancellation of an existing active appointment.		
Format	Button / action link	Valid Range	Available only for active appointment records

Related I/O	REQ_IO601, REQ_IO602
Author	CHIN ZHEN HO

3.1.7 Patient Prescription and Bill Interface

Requirement ID	REQ_IO701	Version	1.0
Item	Patient Prescription List (Output)		
Description	List of prescriptions displayed to the logged-in patient.		
Purpose	To allow a patient to view prescription-related information recorded by the doctor.		
Format	Table	Valid Range	Doctor name, appointment ID, appointment date, appointment time, disease, allergy, prescription
Related I/O	REQ_IO702, REQ_IO703		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO702	Version	1.0
Item	Bill Payment Action (Input)		
Description	Action used by the patient before generating a bill output.		
Purpose	To trigger bill payment handling before bill generation.		
Format	Button	Valid Range	Available only for prescription-related bill action
Related I/O	REQ_IO701, REQ_IO703		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO703	Version	1.0
Item	Bill PDF Output (Output)		

Description	Bill generated in PDF format based on appointment fee and prescription-related information.		
Purpose	To provide a printable and downloadable billing document to the patient.		
Format	PDF document	Valid Range	Single bill generated from valid appointment and prescription data
Related I/O	REQ_IO701, REQ_IO702		
Author	CHIN ZHEN HO		

3.1.8 Doctor Appointment Interface

Requirement ID	REQ_IO801	Version	1.0
Item	Doctor Appointment List (Output)		
Description	List of appointments assigned to the logged-in doctor.		
Purpose	To allow a doctor to review appointments booked by patients.		
Format	Table	Valid Range	Patient details, contact number, appointment date, appointment time, status, available actions
Related I/O	REQ_IO802, REQ_IO803, REQ_IO804		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO802	Version	1.0
Item	Patient Contact Search Field (Input)		
Description	Search field used by the doctor to filter appointment records based on the patient's contact number.		
Purpose	To help the doctor find relevant appointment records more efficiently.		

Format	Text field / search form	Valid Range	Existing patient contact number format
Related I/O	REQ_IO801		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO803	Version	1.0
Item	Doctor Cancel Appointment Action (Input)		
Description	Action available to the doctor for cancelling an appointment.		
Purpose	To allow the doctor to cancel an active appointment when necessary.		
Format	Button / Action link	Valid Range	Available only for existing active appointment records
Related I/O	REQ_IO801		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO804	Version	1.0
Item	Prescribe Action (Input)		
Description	Action available to the doctor to open the prescription entry interface for a selected appointment.		
Purpose	To allow the doctor to create a prescription for a patient appointment.		
Format	Button / action link	Valid Range	Available for appointment records selected for prescription entry
Related I/O	REQ_IO901		
Author	LIM KAI SHEN		

3.1.9 Prescription Entry Interface

Requirement ID	REQ_IO901	Version	1.0
Item	Prescription Entry Form (Input)		
Description	Form used by the doctor to enter prescription information for a selected appointment.		
Purpose	To allow a doctor to record clinical details for a patient.		
Format	Web form	Valid Range	Disease, allergy, prescription text
Related I/O	REQ_IO902, REQ_IO903, REQ_IO904		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO902	Version	1.0
Item	Disease Field (Input)		
Description	Field used by the doctor to record the diagnosed disease or condition.		
Purpose	To store disease-related information in the prescription record.		
Format	Text field / text area	Valid Range	Non-empty text
Related I/O	REQ_IO901		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO903	Version	1.0
Item	Allergy Field (Input)		
Description	Field used by the doctor to record allergy information.		
Purpose	To store patient allergy details in the prescription record.		
Format	Text field / text area	Valid Range	Text
Related I/O	REQ_IO901		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO904	Version	1.0
Item	Prescription Detail Field (Input)		
Description	Field used by the doctor to record prescribed medication or treatment details.		
Purpose	To store prescription instructions for the patient.		
Format	Text area	Valid Range	Non-empty text
Related I/O	REQ_IO901		
Author	PANG TIAN YOU		

3.1.10 Doctor Prescription List Interface

Requirement ID	REQ_IO1001	Version	1.0
Item	Doctor Prescription List (Output)		
Description	List of prescriptions associated with the logged-in doctor's patients.		
Purpose	To allow a doctor to review previously entered prescription records.		
Format	Table	Valid Range	Appointment details, disease, allergy, prescription information
Related I/O	None		
Author	PANG TIAN YOU		

3.1.11 Admin Patient List Interface

Requirement ID	REQ_IO1101	Version	1.0
Item	Patient List View (Output)		
Description	List of all registered patients displayed in the admin dashboard.		
Purpose	To allow the admin to review registered patient records.		

Format	Table	Valid Range	Patient ID, first name, last name, gender, email, contact number
Related I/O	None		
Author	CHIN ZHEN HO		

3.1.12 Admin Doctor List Interface

Requirement ID	REQ_IO1201	Version	1.0
Item	Doctor List View (Output)		
Description	List of doctors displayed in the admin dashboard		
Purpose	To allow the admin to review doctor records stored in the system.		
Format	Table	Valid Range	Username, email, specialization, consultation fee
Related I/O	REQ_IO1301, REQ_1401		
Author	LIM KAI SHEN		

Requirement ID	REQ_IO1202	Version	1.0
Item	Doctor Search Field (Input)		
Description	Search field used by the admin to search doctor records, typically by email.		
Purpose	To allow quick retrieval of doctor information from the admin dashboard.		
Format	Search form	Valid Range	Existing doctor email format
Related I/O	REQ_IO1201		
Author	LIM KAI SHEN		

3.1.13 Add Doctor Interface

Requirement ID	REQ_IO1301	Version	1.0
Item	Add Doctor Form (Input)		
Description	Form used by the admin to create a new doctor account.		
Purpose	To allow the admin to add a doctor into the system.		
Format	Web form	Valid Range	Doctor username, email, password, specialization, consultation fee
Related I/O	REQ_IO1302, REQ_1303		
Author	CHIN ZHEN HO		

Requirement ID	REQ_IO1302	Version	1.0
Item	Doctor Specialization and Fee Fields (Input)		
Description	Fields used to specify the new doctor's specialization and consultation fee.		
Purpose	To store doctor profile information required for appointment booking and display.		
Format	Text field/ number field	Valid Range	Valid specialization text and positive numeric consultation fee
Related I/O	REQ_IO1301		
Author	CHIN ZHEN HO		
Requirement ID	REQ_IO1303	Version	1.0
Item	Doctor Account Credentials (Input)		
Description	Login credentials created for the new doctor account.		
Purpose	To enable the added doctor to access the doctor dashboard.		
Format	Text field, email field, password field	Valid Range	Unique username or email according to system data;

			non-empty password
Related I/O	REQ_IO1301		
Author	CHIN ZHEN HO		

3.1.14 Delete Doctor Interface

Requirement ID	REQ_IO1401	Version	1.0
Item	Delete Doctor Action (Input)		
Description	Interface used by the admin to remove a doctor record from the system.		
Purpose	To allow deletion of an existing doctor account.		
Format	Action link / button / delete form	Valid Range	Existing doctor record only
Related I/O	REQ_IO1201		
Author	PANG TIAN YOU		

3.1.15 Admin Appointment Detail Interface

Requirement ID	REQ_IO1501	Version	1.0
Item	Appointment Detail View (Output)		
Description	Detailed list of appointment records displayed in the admin dashboard.		
Purpose	To allow the admin to review all appointments recorded in the system.		
Format	Table	Valid Range	Patient details, doctor name, consultation fee, appointment date, appointment time, patient status, doctor status
Related I/O	REQ_IO1502		

Author	PANG TIAN YOU
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Requirement ID	REQ_IO1502	Version	1.0
Item	Appointment Search Field (Input)		
Description	Search field used by the admin to filter appointment records, typically by patient contact number.		
Purpose	To allow efficient retrieval of appointment information.		
Format	Search form	Valid Range	Existing patient contact number format
Related I/O	REQ_IO1501		
Author	PANG TIAN YOU		

3.1.16 Admin Prescription List Interface

Requirement ID	REQ_IO1601	Version	1.0
Item	Admin Prescription List View (Output)		
Description	List of prescription records displayed in the admin dashboard.		
Purpose	To allow the admin to review prescription and appointment-related details.		
Format	Table	Valid Range	Doctor, patient, appointment ID, appointment date, appointment time, disease, allergy, prescription
Related I/O	None		
Author	LIM KAI SHEN		

3.1.17 Query Management Interface

Requirement ID	REQ_IO1701	Version	1.0
Item	User Query List View (Output)		

Description	List of contact messages or feedback records displayed in the admin dashboard.		
Purpose	To allow the admin to review queries submitted through the system contact page.		
Format	Table	Valid Range	Name, email, contact number, message
Related I/O	REQ_IO1702		
Author	PANG TIAN YOU		

Requirement ID	REQ_IO1702	Version	1.0
Item	Query Search Field (Input)		
Description	Search field used by the admin to search submitted query records.		
Purpose	To allow efficient retrieval of user-submitted messages.		
Format	Search form	Valid Range	Existing contact number or related search value
Related I/O	REQ_IO1701		
Author	PANG TIAN YOU		

3.2 Functional Requirements

This section describes the functional requirements of the Hospital Management System, including the overall system requirements and the specific features available to different users. Figure 2.0 shows the overall use case diagram of the system.

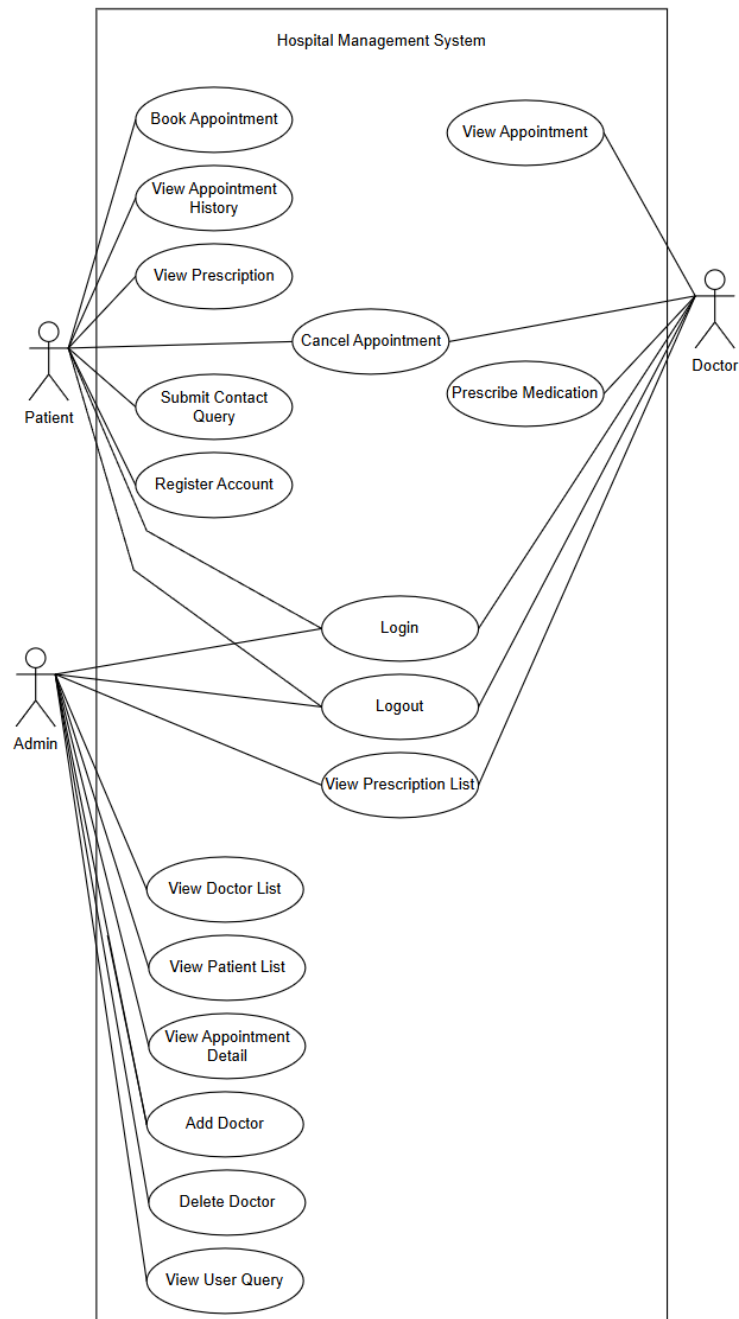


Figure 2.0 Hospital Management System Overall Use Case

3.2.1 User Authentication

3.2.1.1 F001 Register Account

Requirement ID	REQ_F001
Version	1.0
Description	The system shall allow patients to register a new account by providing personal details such as name, email, contact number, password, and gender.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC001
Version	1.0
Use Case	F001 Register Account
Purpose	To allow a patient to create a new account.
Actor	Patient
Trigger	The patient opens the home page and enters the required registration details.
Precondition	The patient is not logged into the system and the registration form is available on the home page.
Main Flow	1. The patient accesses the home page. 2. The system displays the patient registration form. 3. The patient enters first name, last name, email, phone number, password, confirm password, and choose gender. 4. The patient clicks the Register button. 5. The system validates the entered information. 6. The system creates a new patient account. 7. The system redirects the patient to the patient dashboard.
Alternate Flow	1. If any required field is empty, the system displays a validation message.

	2. If the password and confirm password do not match, the system displays “Not Matching.” 3. If the password is fewer than 6 characters, the system displays an error message.
Rules	All required fields must be completed. The password and confirm password must match. The password must contain at least 6 characters.
Author	CHIN ZHEN HO

Table 5.0 Use Case Specification – Register Account

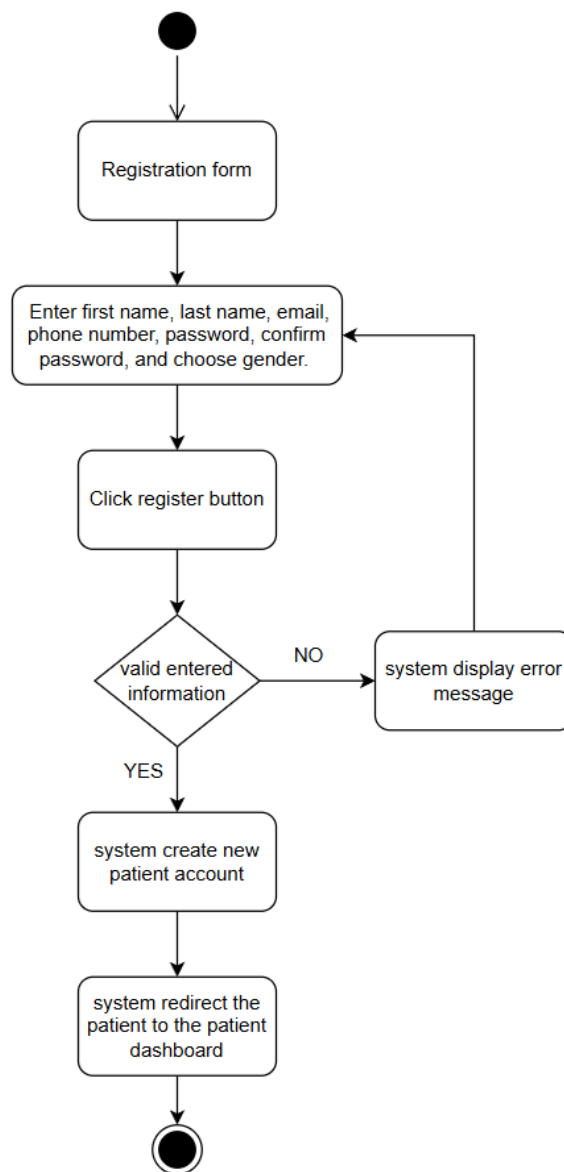


Figure 3.0 State diagram - Register Account

3.2.1.2 F002 Login

Requirement ID	REQ_F002
Version	1.0
Description	The system shall allow patients, doctors, and administrators to log in using valid credentials (email/username and password).
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC002
Version	1.0
Use Case	F002 Login
Purpose	To allow patients, doctors, and administrators to log into the system using valid credentials.
Actor	Patient, Doctor, Admin
Trigger	The user accesses the login page and enters valid login credentials.
Precondition	The user is not logged into the system and a valid account already exists.
Main Flow	1. The user accesses the home page. 2. The system displays the available login options for Patient, Doctor, and Admin. 3. For a patient, the user may click “Already have an account?” from the patient registration page to open the patient login page. 4. For a doctor, the user selects the Doctor tab to open the doctor login page. 5. For admin, the user selects the Receptionist tab to open the admin login page. 6. The user enters the required login credentials. 7. The user clicks the Login button.

	8. The system validates the entered credentials. 9. The system grants access and redirects the user to the corresponding dashboard.
Alternate Flow	1. If the username, email, or password field is empty, the system displays a validation message. 2. If the entered credentials are invalid, the system denies login and displays an error message.
Rules	1. A valid account must exist before login. 2. The user must enter the correct credentials for the selected role. 3. The system shall redirect the user to the correct dashboard after successful login.
Author	CHIN ZHEN HO

Table 6.0 Use Case Specification – Login

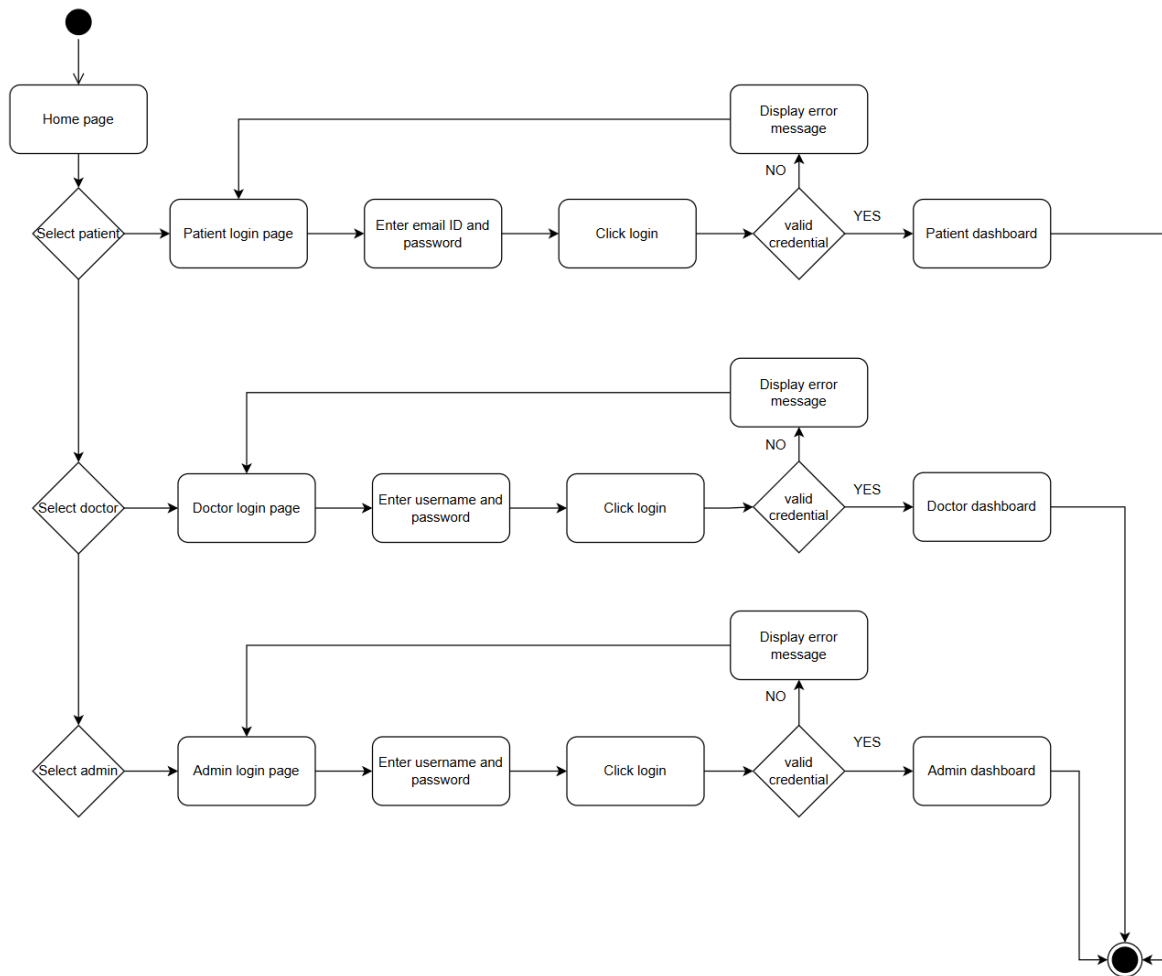


Figure 4.0 State diagram - Login

3.2.1.3 F003 Logout

Requirement ID	REQ_F003
Version	1.0
Description	The system shall allow users to log out from their account and terminate the current session.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC003
Version	1.0
Use Case	F003 Logout
Purpose	To allow users to exit the system.

Actor	Patient, Doctor, Admin
Trigger	User click Logout button from the corresponding dashboard.
Precondition	User is logged into the system and is currently on the patient, doctor, or administrator dashboard.
Main Flow	1. The user clicks the Logout button. 2. The system terminate current user session. 3. The system redirects user to the logout page. 4. The system display message "You have logged out." 5. The system display button labeled "Back to Login Page". 6. The user click button to return to the login page.
Alternate Flow	—
Rules	The system must terminate the user session after logout.
Author	CHIN ZHEN HO

Table 7.0 Use Case Specification – Logout

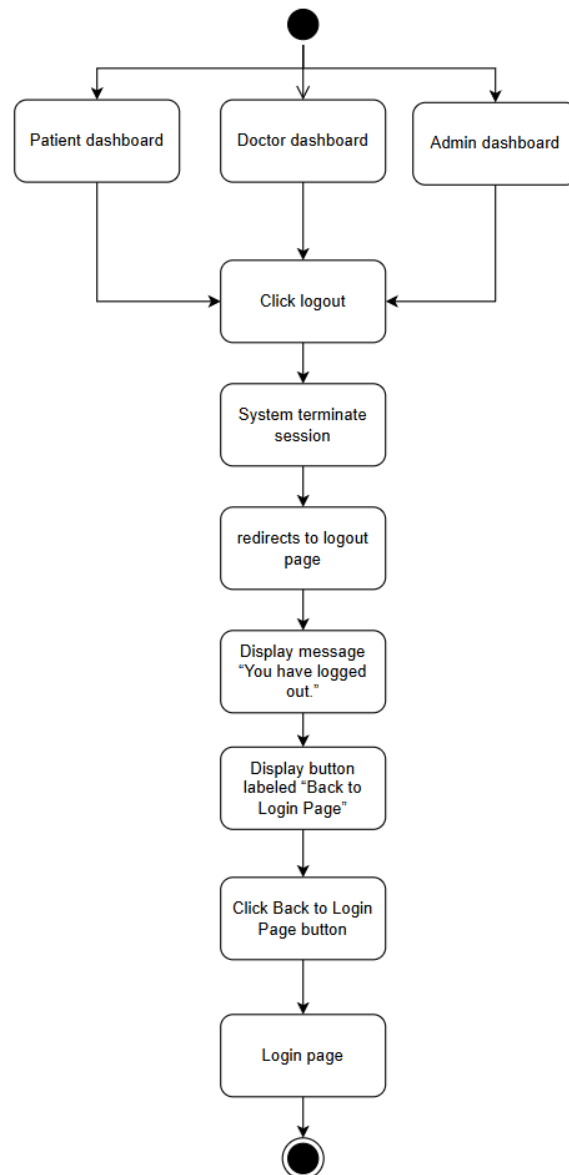


Figure 5.0 State diagram - Logout

3.2.2 Patient Functions

3.2.2.1 F004 Book Appointment

Requirement ID	REQ_F004
Version	1.0
Description	The system shall allow patients to book an appointment by selecting a specialization, choosing a doctor, selecting a date and time, and viewing the consultation fee before confirming the booking.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC004
Version	1.0
Use Case	F004 Book Appointment
Purpose	To allow patients to book appointments.
Actor	Patient
Trigger	Patient selects the Book Appointment function from the dashboard.
Precondition	Logged in.
Main Flow	1. Patient select Book Appointment. 2. System display appointment booking page. 3. Patient select specialization. 4. Patient select doctor. 5. Patient select date. 6. Patient select time. 7. Patient click the Create new entry button. 8. The system validate the entered appointment information. 9. The system create a new appointment record. 10. The system display the message “Your appointment successfully booked”.

Alternate Flow	1. If the selected doctor is not available on the chosen date or time, the system displays the message “We are sorry to inform that the doctor is not available in this time or date. Please choose different time or date!” 2. If the selected appointment date or time is not in the future, the system displays the message “Select a time or date in the future!” 3. If any required field is not selected, the system displays a validation message
Rules	1. The patient must be logged into the system. 2. The patient must select a specialization, doctor, appointment date, and appointment time. 3. The consultation fee shall be displayed based on the selected doctor. 4. The appointment date and time must be in the future. 5. The selected doctor must be available at the chosen date and time.
Author	CHIN ZHEN HO

Table 8.0 Use Case Specification – Book Appointment

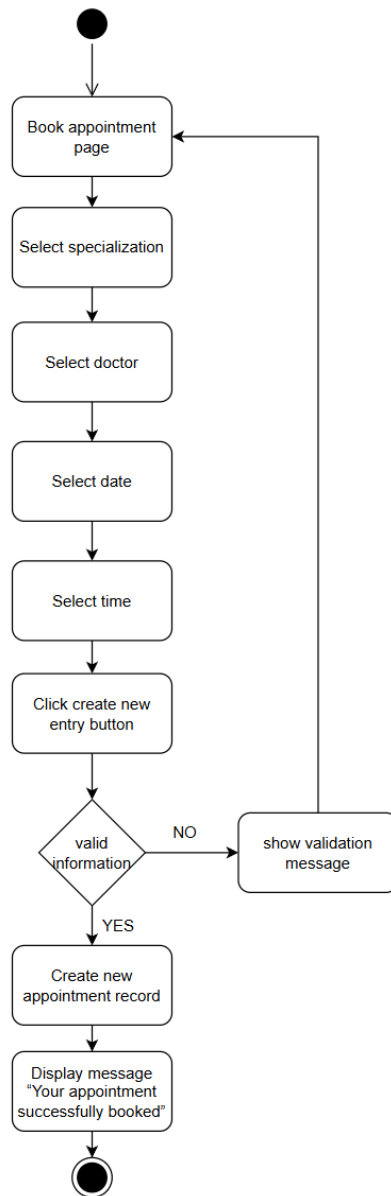


Figure 6.0 State diagram – Book Appointment

3.2.2.2 F005 View Appointment History

Requirement ID	REQ_F005
Version	1.0
Description	The system shall allow patients to view their appointment history including doctor name, consultation fee, appointment date and time, and current status (e.g., Active, Cancelled).
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC005
Version	1.0
Use Case	F005 View Appointment History
Purpose	To allow patients to view past appointment.
Actor	Patient
Trigger	The patient selects the Appointment History function from the dashboard.
Precondition	The patient is logged into the system and has existing appointment records.
Main Flow	1. The patient selects the Appointment History function. 2. The system retrieves the patient's appointment records. 3. The system displays the appointment history list. 4. The patient views appointment details including doctor name, consultation fees, appointment date, appointment time, and current status.
Alternate Flow	1. If there are no appointment records, the system displays an empty list or appropriate message. 2. If the appointment status is Active, the patient may choose to cancel the appointment by clicking the Cancel button.
Rules	1. The patient must be logged into the system.

	2. Only appointments belonging to the logged-in patient shall be displayed. 3. The system shall display the current status of each appointment. 4. Only Active appointments can be cancelled.
Author	CHIN ZHEN HO

Table 9.0 Use Case Specification – View Appointment History

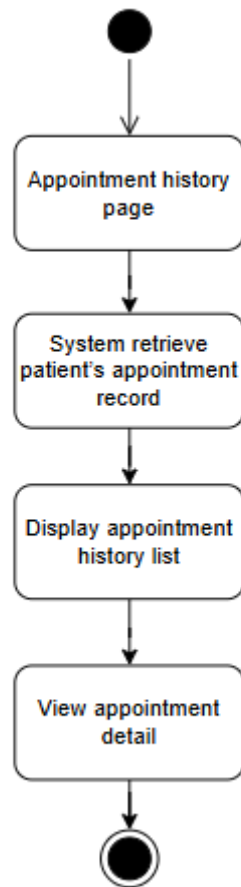


Figure 7.0 State diagram – View Appointment History

3.2.2.3 F006 Cancel Appointment

Requirement ID	REQ_F006
Version	1.0
Description	The system shall allow patients to cancel their existing appointments.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC006
Version	1.0
Use Case	F006 Cancel Appointment
Purpose	To allow a patient to cancel an active appointment.
Actor	Patient
Trigger	Patient click Cancel button for an active appointment.
Precondition	The patient is logged into the system and has at least one active appointment.
Main Flow	1. The patient selects the Appointment History function. 2. The system displays the patient's appointment history. 3. The patient click the Cancel button for an active appointment. 4. The system displays the confirmation message "Are you sure you want to cancel this appointment?" 5. The patient click OK to confirm the cancellation. 6. The system updates the appointment status to cancelled. 7. The system displays the message "Your appointment successfully cancelled". 8. The system redirects the patient back to the dashboard page.
Alternate Flow	1. If the patient clicks Cancel in the confirmation dialog, the system does not cancel the appointment and returns to the appointment history page.

	2. If the selected appointment is already cancelled, the system does not allow the cancellation process.
Rules	1. The patient must be logged into the system. 2. Only active appointments can be cancelled. 3. The system shall request confirmation before cancelling the appointment.
Author	CHIN ZHEN HO

Table 10.0 Use Case Specification – Cancel Appointment

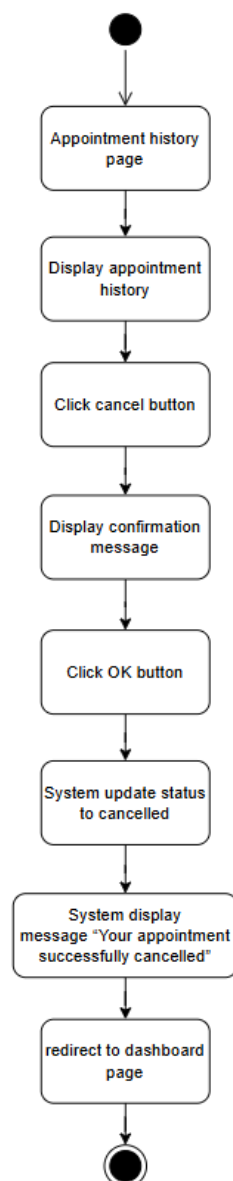


Figure 8.0 State diagram – Cancel Appointment

3.2.2.4 F007 View Prescription

Requirement ID	REQ_F007
Version	1.0
Description	The system shall allow patients to view prescriptions including disease, allergies, and medication details. The system shall also allow patients to make bill payments and generate a printable bill in PDF format after payment.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC007
Version	1.0
Use Case	F007 View Prescription
Purpose	To allow a patient to view prescription details, make bill payment, and generate a bill in PDF format.
Actor	Patient
Trigger	Patient select Prescriptions function from the dashboard.
Precondition	Patient is logged into the system and has at least one prescription record available.
Main Flow	1. The patient selects the Prescriptions function. 2. The system retrieves the patient's prescription records. 3. The system displays prescription details including doctor name, appointment ID, appointment date, appointment time, diseases, allergies, and prescribed medication. 4. The patient clicks the Pay Bill button. 5. The system processes the bill payment. 6. The system displays the message "Bill Paid Successfully". 7. The patient clicks OK.

	8. The system generates and displays the bill in PDF format.
Alternate Flow	1. If there are no prescription records, the system displays an empty list. 2. If the bill payment is not completed successfully, the system does not generate the PDF bill.
Rules	1. The patient must be logged into the system. 2. Only prescription records belonging to the logged-in patient shall be displayed. 3. The bill must be paid before the PDF bill is generated.
Author	CHIN ZHEN HO

Table 11.0 Use Case Specification – View Prescription

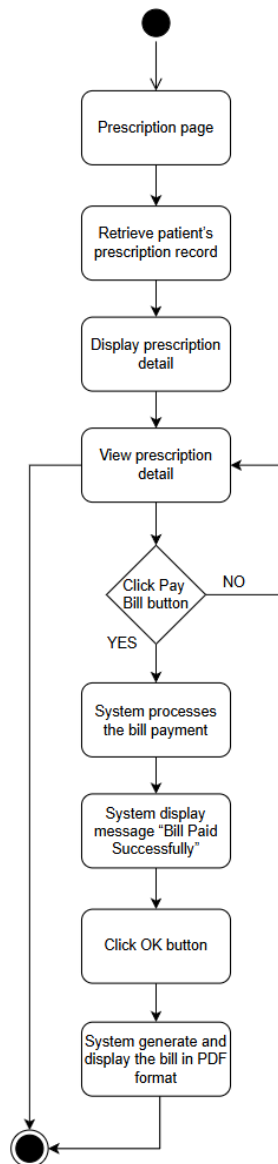


Figure 9.0 State diagram – View Prescription

3.2.2.5 F008 Submit Contact Query

Requirement ID	REQ_F008
Version	1.0
Description	The system shall allow patient to submit queries or feedback through the contact form.
Author	CHIN ZHEN HO

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC008
Version	1.0
Use Case	F008 Submit Contact Query
Purpose	To submit feedback or query through contact page.
Actor	Patient
Trigger	Open the Contact page and click the Send Message button.
Precondition	contact form is available on the website.
Main Flow	1. Open the Contact page. 2. System displays the contact form. 3. Enter name, email, phone number, and message. 4. Click the Send Message button. 5. System validates the entered information. 6. System stores the submitted query or feedback.
Alternate Flow	1. If any required field is empty, the system display validation message. 2. If the entered information is invalid, the system does not submit the form.
Rules	1. All required fields must be completed before submission. 2. The contact form must contain name, email, phone number, and message.
Author	CHIN ZHEN HO

Table 12.0 Use Case Specification – Submit Contact Query

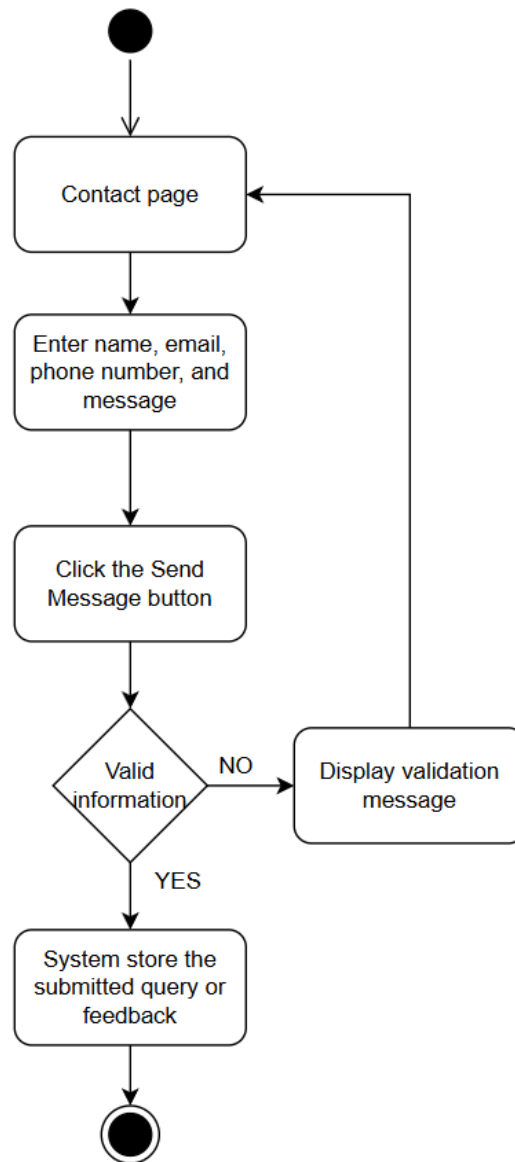


Figure 10.0 State diagram – Submit Contact Query

3.2.3 Doctor Functions

3.2.3.1 F009 View Appointment

Requirement ID	REQ_F009
Version	1.0
Description	The system shall allow doctors to view appointments booked by patients, including patient details, appointment date and time, and current status.
Author	PANG TIAN YOU

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC009
Version	1.0
Use Case	F009 View Appointment
Purpose	To allow doctors to view appointments assigned to them.
Actor	Doctor
Trigger	The doctor selects the Appointments function from the doctor dashboard.
Precondition	The doctor is logged into the system and the doctor dashboard is available.
Main Flow	1. The doctor selects the Appointment function. 2. The system retrieves appointment records assigned to the logged-in doctor. 3. The system displays the appointment list, including patient details, contact number, appointment date, appointment time, and current status. 4. The doctor reviews the appointment information. 5. The system keeps the appointment list available for further doctor actions such as search, cancellation, or prescription entry.
Alternate Flow	1. If there are no appointments assigned to the doctor, the system displays an empty list or an appropriate message.

	2. If the doctor enters a patient's contact number in the search field, the system filters and displays matching appointment records. 3. If the search query does not match any appointment, the system displays an appropriate message and remains on the appointment list page.
Rules	1. Only appointments assigned to the logged-in doctor shall be displayed. 2. Appointment search shall be performed using the patient contact number. 3. Appointment information shall be displayed in read-only form unless the doctor selects another available action.
Author	PANG TIAN YOU

Table 13.0 Use Case Specification – View Appointment

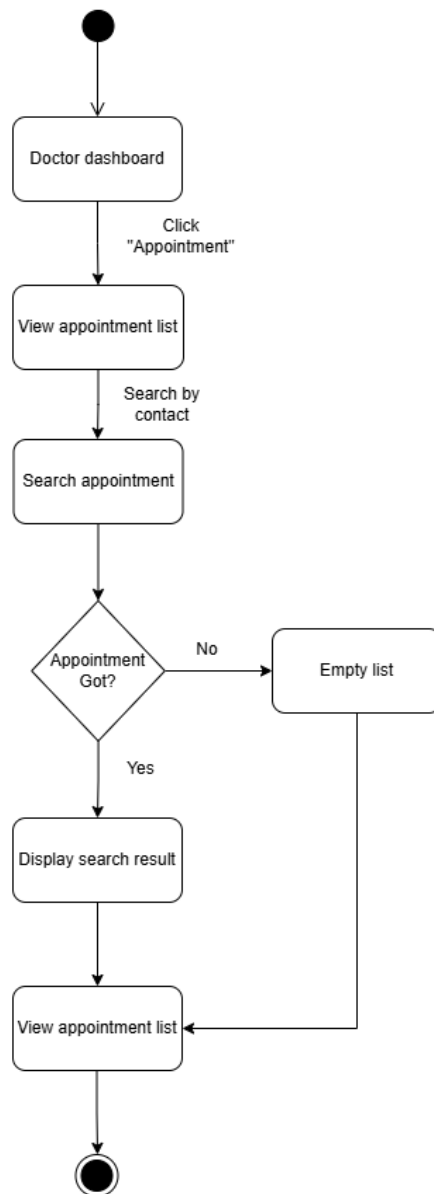


Figure 11.0 State diagram – View Appointment

3.2.3.2 F010 Prescribe Medication

Requirement ID	REQ_F010
Version	1.0
Description	The system shall allow doctors to enter and submit prescription details including disease, allergies, and medication for a selected appointment.
Author	PANG TIAN YOU

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC010
Version	1.0
Use Case	F010 Prescribe Medication
Purpose	To allow doctors to create a prescription for a selected patient appointment.
Actor	Doctor
Trigger	The doctor clicks the Prescribe action for a selected appointment.
Precondition	The doctor is logged into the system, the selected appointment exists, and the appointment is active.
Main Flow	1. The doctor opens the Appointment page. 2. The system displays the list of appointments assigned to the doctor. 3. The doctor selects the Prescribe action for a chosen appointment. 4. The system displays the prescription entry form. 5. The doctor enters disease, allergy, and prescription details. 6. The doctor clicks the Submit button. 7. The system validates the entered prescription information. 8. The system stores the prescription record linked to the selected appointment and patient. 9. The system displays a success message indicating that the prescription has been recorded successfully.

	10. The system redirects the doctor to the relevant doctor page for continued operation.
Alternate Flow	1. If any required field is empty, the system displays a validation message and does not save the prescription. 2. If the selected appointment is invalid or no longer active, the system does not create the prescription and returns to the appointment list. 3. If the prescription record cannot be stored successfully, the system displays an error message, and no prescription record is created.
Rules	1. A prescription shall be linked to a selected appointment record. 2. Only the logged-in doctor shall be allowed to create a prescription for appointments assigned to that doctor. 3. Disease and prescription details must be entered before submission.
Author	PANG TIAN YOU

Table 14.0 Use Case Specification – Prescribe Medication

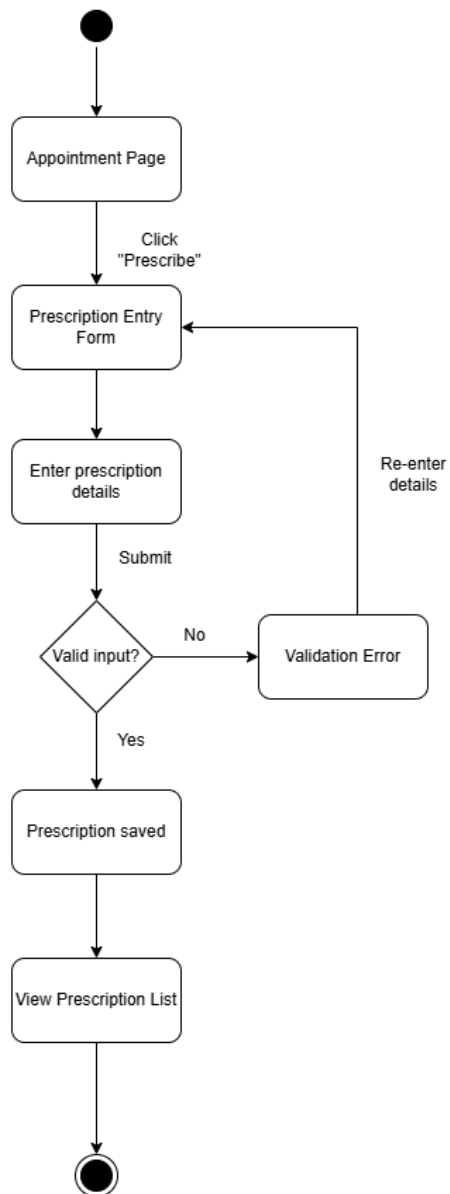


Figure 12.0 State diagram – Prescribe Medication

3.2.3.3 F011 Cancel Appointment

Requirement ID	REQ_F011
Version	1.0
Description	The system shall allow doctors to cancel appointments with patients.
Author	PANG TIAN YOU

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC011
Version	1.0
Use Case	F011 Cancel Appointment
Purpose	To allow a doctor to cancel an active appointment assigned to them.
Actor	Doctor
Trigger	The doctor clicks the Cancel action for an active appointment.
Precondition	The doctor is logged into the system and the selected appointment record exists.
Main Flow	1. The doctor selects the Appointment function from the doctor dashboard. 2. The system displays the appointment list assigned to the doctor. 3. The doctor clicks the Cancel action for an active appointment. 4. The system displays a confirmation prompt asking whether the doctor wants to cancel the appointment. 5. The doctor confirms the cancellation. 6. The system updates the appointment status to indicate that the appointment has been cancelled by the doctor. 7. The system displays a success message confirming the cancellation. 8. The system refreshes the appointment list and displays the updated appointment status.

Alternate Flow	1. If the doctor cancels the confirmation prompt, the system does not update the appointment and returns to the appointment list. 2. If the selected appointment has already been cancelled, the system does not perform the cancellation process and displays an appropriate message. 3. If the appointment status update fails, the system displays an error message and keeps the original appointment status unchanged.
Rules	1. Only active appointments can be cancelled by the doctor. 2. The doctor shall only be allowed to cancel appointments assigned to that doctor. 3. Appointment records shall be updated through status change rather than physical deletion from the database.
Author	PANG TIAN YOU

Table 15.0 Use Case Specification – Cancel Appointment

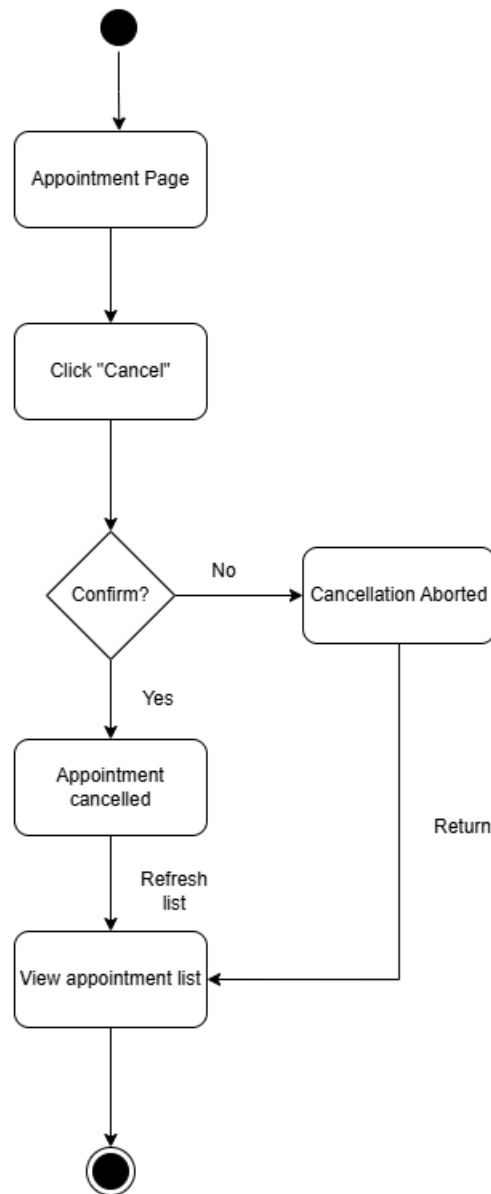


Figure 13.0 State diagram – Cancel Appointment

3.2.3.4 F012 View Prescription List

Requirement ID	REQ_F012
Version	1.0
Description	The system shall allow doctors to view the list of prescriptions associated with their patients, including appointment details, diseases, allergies, and prescribed medications.
Author	PANG TIAN YOU

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC012
Version	1.0
Use Case	F012 View Prescription List
Purpose	To allow doctors to review prescription records associated with their patients.
Actor	Doctor
Trigger	The doctor selects the Prescription List function from the doctor's dashboard.
Precondition	The doctor is logged into the system and the prescription list page is available.
Main Flow	1. The doctor selects the Prescription List function. 2. The system retrieves prescription records associated with patients assigned to the logged-in doctor. 3. The system displays the prescription list, including appointment details, disease, allergy, and prescription information. 4. The doctor reviews the prescription records.
Alternate Flow	1. If there are no prescription records associated with the doctor's patients, the system displays an empty list or an appropriate message.

Rules	1. Only prescription records associated with the logged-in doctor's patients shall be displayed. 2. Prescription information in this view shall be displayed in read-only form.
Author	PANG TIAN YOU

Table 16.0 Use Case Specification – View Prescription List

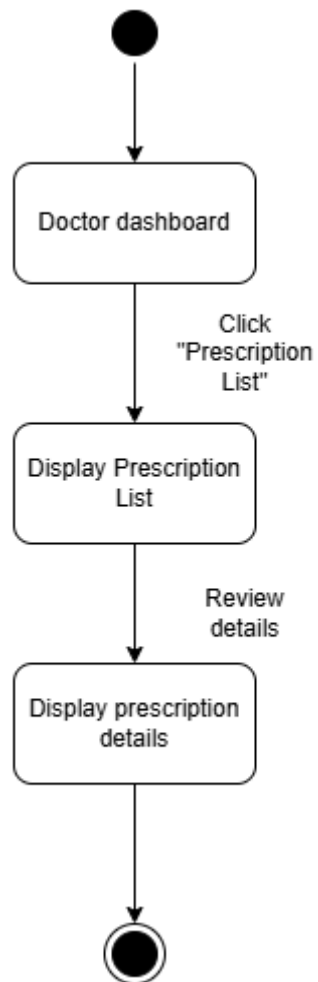


Figure 14.0 State diagram – View Prescription List

3.2.4 Admin Functions

3.2.4.1 F013 Add Doctor

Requirement ID	REQ_F013
Version	1.0
Description	The system shall allow the admin to add a new doctor by providing details such as name, specialization, email, password, and consultancy fees.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC013
Version	1.0
Use Case	F013 Add Doctor
Purpose	To allow an admin to register a new doctor into the hospital management system.
Actor	Admin
Trigger	The admin navigates to the "Add Doctor" section and submits the required details.
Precondition	The admin is successfully logged into the dashboard.
Main Flow	1. The admin selects the "Add Doctor" tab.

	2. The admin enters the Doctor Name, Specialization, Email ID, Password, Confirm Password, and Consultancy Fees. 3. The admin clicks "Add Doctor". 4. The system validates the input. 5. The system creates a new doctor record. 6. The system displays an alert: "Doctor added successfully!".
Alternate Flow	1. If "Password" and "Confirm Password" do not match, a real-time UI indicator shows "Not Matching". 2. If inputs are invalid, the database insertion fails and the process aborts.
Rules	All fields are mandatory. Passwords must match. Name input is restricted to alphabetic characters via JavaScript.
Author	LIM KAI SHEN

Table 17.0 Use Case Specification – Add Doctor

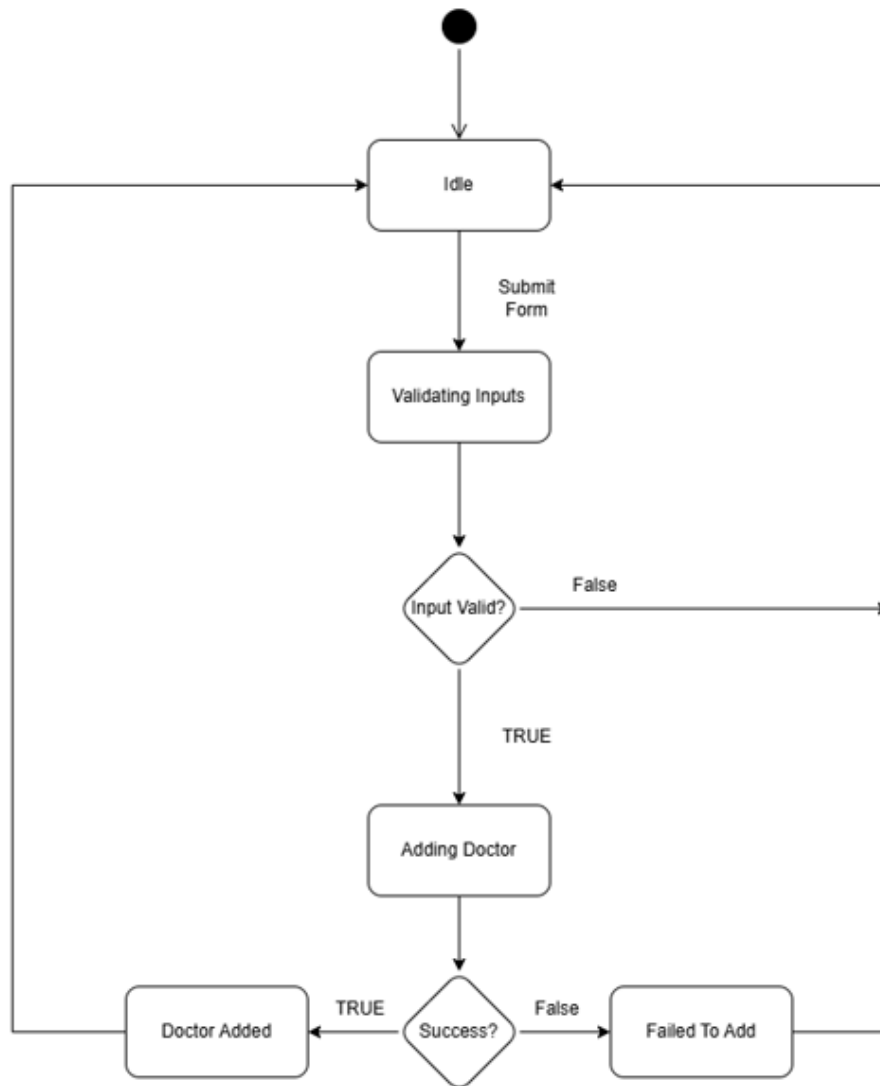


Figure 15.0 State diagram – Add Doctor

3.2.4.2 F014 Delete Doctor

Requirement ID	REQ_F014
Version	1.0
Description	The system shall allow the admin to remove an existing doctor from the system using their registered email address.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC014
Version	1.0
Use Case	F014 Delete Doctor
Purpose	To remove a doctor's access and records from the active hospital management system.
Actor	Admin
Trigger	The admin navigates to the "Delete Doctor" section and submits the doctor's email ID.
Precondition	The admin is logged in, and the target doctor's email exists in the database.
Main Flow	1. The admin selects the "Delete Doctor" tab. 2. The admin enters the target Email ID. 3. The admin clicks "Delete Doctor". 4. The system prompts for confirmation.

	5. The admin confirms. 6. The system deletes the record and displays: "Doctor removed successfully!".
Alternate Flow	1. If the admin denies the confirmation prompt, the action aborts. 2. If the deletion query fails, the system alerts: "Unable to delete!".
Rules	The email ID must belong to an existing doctor in doctb.
Author	LIM KAI SHEN

Table 18.0 Use Case Specification – Delete Doctor

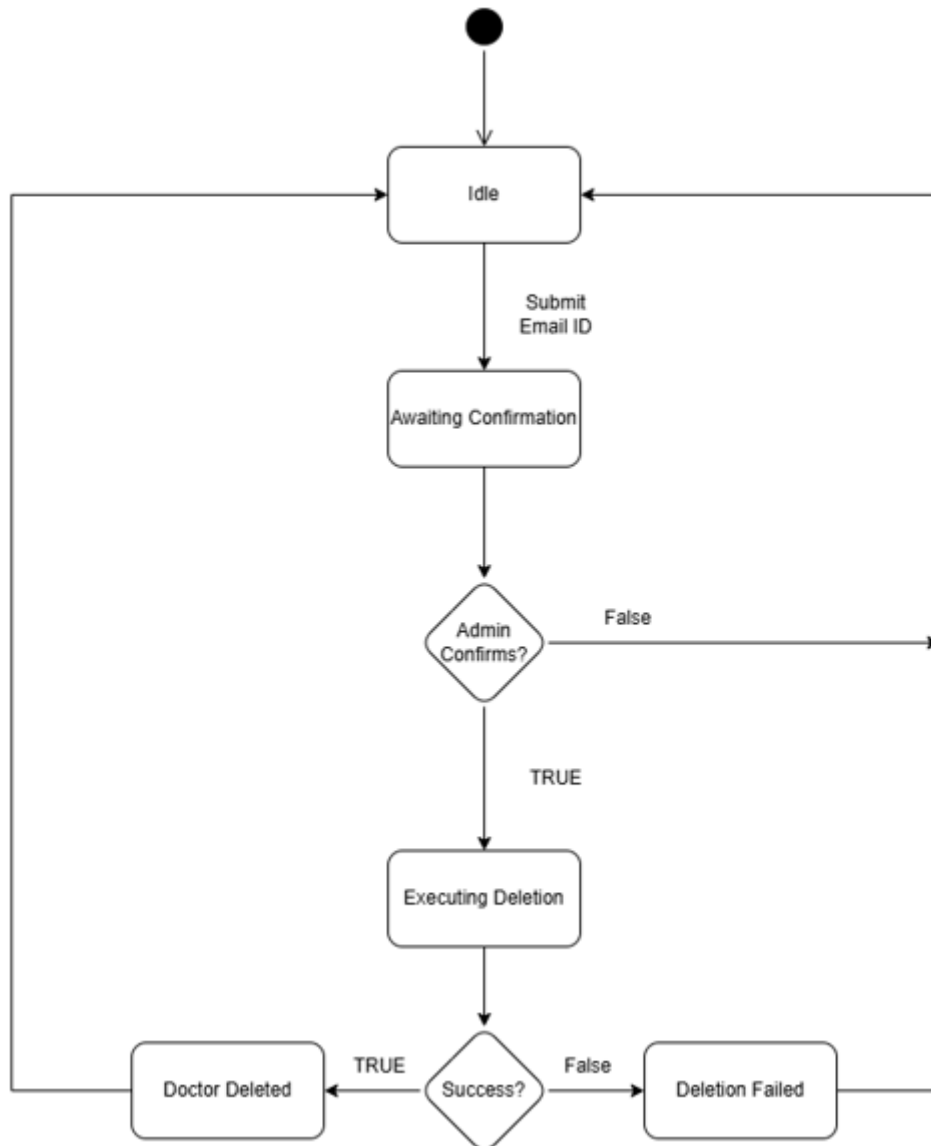


Figure 16.0 State diagram – Delete Doctor

3.2.4.3 F015 View & Search Doctor List

Requirement ID	REQ_F015
Version	1.0
Description	The system shall display a list of all registered doctors and allow the admin to search for a specific doctor by Email ID.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC015
Version	1.0
Use Case	F015 View & Search Doctor List
Purpose	To allow the admin to monitor and search through the system's registered medical professionals.
Actor	Admin
Trigger	The admin clicks the "Doctor List" tab or submits a search query.
Precondition	The admin is logged in.
Main Flow	1. The admin selects the "Doctor List" tab. 2. The system retrieves all doctor records from the database. 3. The system displays the records (Name, Specialization, Email, Password, Fees) in a tabular format.

	4. The admin enters an Email ID into the search bar and clicks "Search". 5. The system fetches and displays the specific doctor's details.
Alternate Flow	1. If the search query does not match any database record, the system alerts: "No entries found!" and redirects back to the full list.
Rules	The search input must specifically target the doctor's registered Email ID.
Author	LIM KAI SHEN

Table 19.0 Use Case Specification – View & Search Doctor List

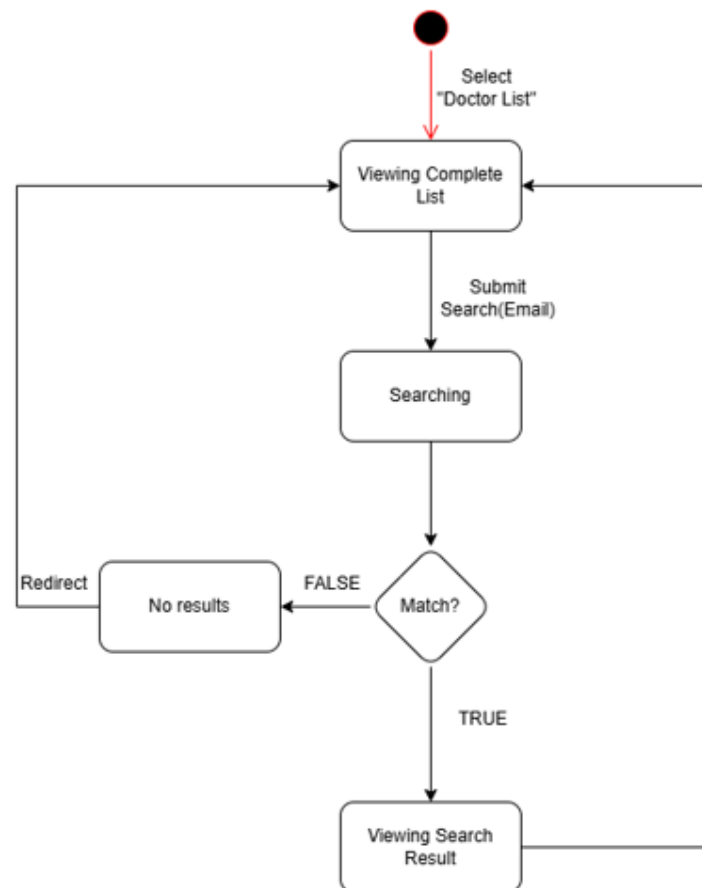


Figure 17.0 State diagram – View & Search Doctor List

3.2.4.4 F016 View & Search Patient List

Requirement ID	REQ_F016
Version	1.0
Description	The system shall display a list of all registered patients and allow the admin to search for a specific patient by their Contact number.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC016
Version	1.0
Use Case	F016 View & Search Patient List
Purpose	To allow the admin to monitor all registered patient accounts.
Actor	Admin
Trigger	The admin clicks the "Patient List" tab or submits a search query.
Precondition	The admin is logged in.
Main Flow	1. The admin selects the "Patient List" tab. 2. The system retrieves and displays all patient records (ID, Name, Gender, Email, Contact, Password) in a table.

	3. The admin inputs a Contact number into the search bar and clicks "Search". 4. The system fetches and displays the matching patient's details.
Alternate Flow	1. If no patient matches the contact number, the system alerts: "No entries found! Please enter valid details" and returns to the dashboard.
Rules	Search queries are executed based strictly on the patient's exact contact number.
Author	LIM KAI SHEN

Table 20.0 Use Case Specification – View & Search Patient List

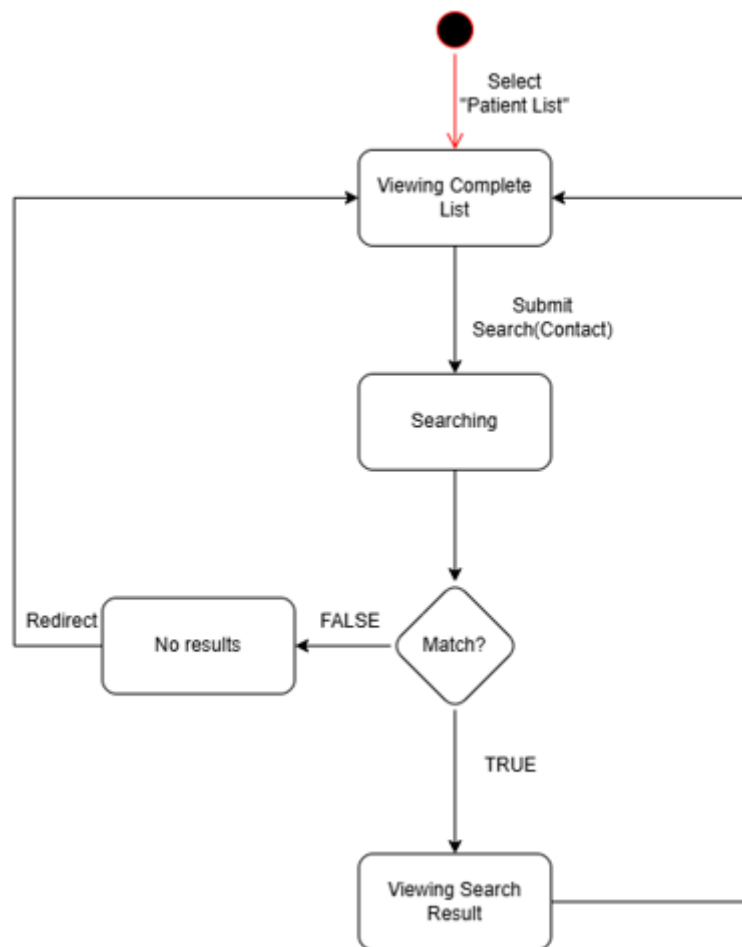


Figure 18.0 State diagram – View & Search Patient List

3.2.4.5 F017 View & Search Appointment Details

Requirement ID	REQ_F017
Version	1.0
Description	The system shall display a comprehensive list of all system-wide appointments and allow the admin to filter them by the patient's Contact number.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC017
Version	1.0
Use Case	F017 View & Search Appointment Details
Purpose	To provide the admin with oversight of all scheduled, active, and cancelled hospital appointments.
Actor	Admin
Trigger	The admin clicks the "Appointment Details" tab.
Precondition	The admin is logged in.
Main Flow	1. The admin selects the "Appointment Details" tab.

	2. The system displays all appointments, including status (Active, Cancelled by Patient, Cancelled by Doctor). 3. The admin searches by Contact number. 4. The system returns the specific appointment history associated with that contact.
Alternate Flow	1. If the contact number yields no appointment records, the system displays an error alert and redirects back to the dashboard.
Rules	Appointment status is determined dynamically based on the userStatus and doctorStatus database flags.
Author	LIM KAI SHEN

Table 20.0 Use Case Specification – View & Search Appointment Details

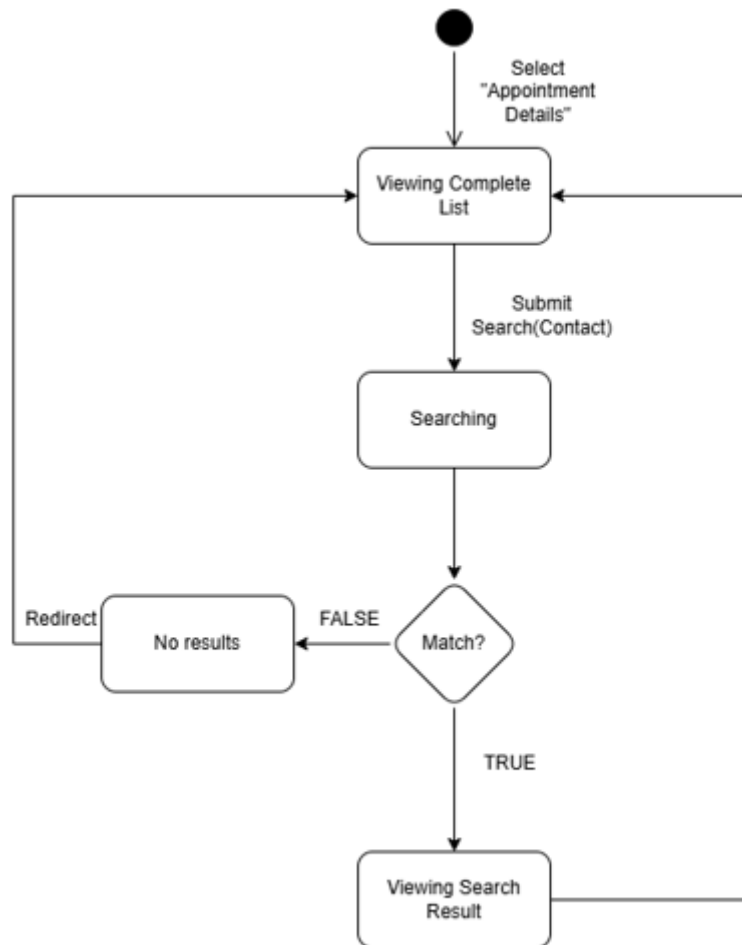


Figure 19.0 State diagram – View & Search Appointment Details

3.2.4.6 F018 View Prescription List

Requirement ID	REQ_F018
Version	1.0
Description	The system shall display a list of all medical prescriptions issued by doctors to patients.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC018
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Version	1.0
Use Case	F018 View Prescription List
Purpose	To maintain an administrative audit trail of all diagnoses and prescriptions issued within the hospital.
Actor	Admin
Trigger	The admin clicks the "Prescription List" tab.
Precondition	The admin is logged in.
Main Flow	1. The admin selects the "Prescription List" tab. 2. The system retrieves all records from the prestb database table. 3. The system displays the Doctor Name, Patient Details, Appointment details, Disease, Allergies, and Prescription text in a table.
Alternate Flow	N/A (Standard data retrieval).
Rules	The displayed data is read-only for the administrator.
Author	LIM KAI SHEN

Table 21.0 Use Case Specification – View Prescription List

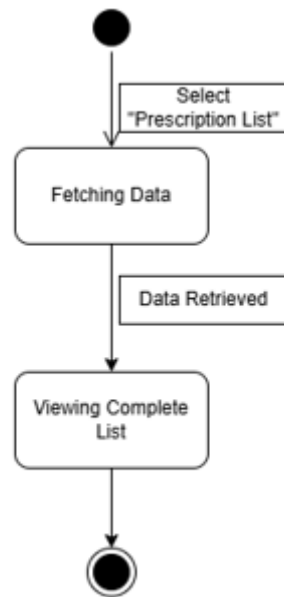


Figure 20.0 State diagram – View Prescription List

3.2.4.7 F019 View & Search User Queries

Requirement ID	REQ_F019
Version	1.0
Description	The system shall allow the admin to view messages and queries submitted through the public Contact Us page.
Author	LIM KAI SHEN

The following table shows the details of this feature, followed by a state diagram to illustrate the state transitions of the process.

Use Case ID	UC019
Version	1.0
Use Case	F019 View & Search User Queries
Purpose	To enable the admin to review feedback or inquiries submitted by the public.
Actor	Admin
Trigger	The admin clicks the "Queries" tab.
Precondition	The admin is logged in.
Main Flow	1. The admin selects the "Queries" tab. 2. The system displays all submitted messages (Name, Email, Contact, Message). 3. The admin searches by Contact number. 4. The system filters and displays matching queries.

Alternate Flow	1. If the search fails to match a contact number, an alert is shown indicating no entries were found.
Rules	Data is fetched from the contact database table.
Author	LIM KAI SHEN

Table 22.0 Use Case Specification – View & Search User Queries

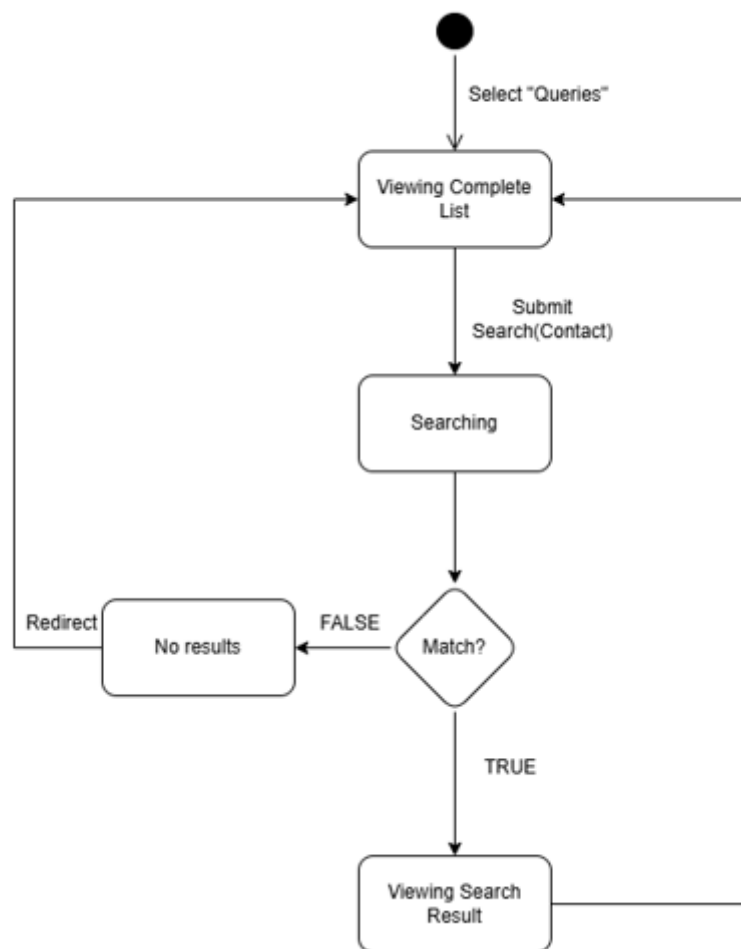


Figure 21.0 State diagram – View & Search User Queries

3.3 Performance Requirement

The performance requirement for HMS shown as below:

Requirement ID	Description	Priority	Author
REQ_PR001	HMS shall generate pages and respond to user input in 5 seconds or less for given typical local deployment conditions.	High	CHIN ZHEN HO
REQ_PR002	HMS shall complete user operations such as login, appointment booking, search, etc., within 5 seconds under normal local deployment conditions.	High	PANG TIAN YOU
REQ_PR003	HMS shall generate a single bill PDF within 10 seconds or less for given typical local deployment conditions.	Medium	LIM KAI SHEN

3.4 Design Constraint

The development, execution, and maintenance of the Hospital Management System (HMS) are limited by the following design and implementation constraints:

3.4.1 Technology Stack Limitations

- **Frontend:** The user interface is strictly constrained to HTML5, CSS3, and JavaScript. All responsive design and layout components must utilize the Bootstrap 4 framework. Modern reactive frontend frameworks (e.g., React, Vue.js) are not utilized.
- **Backend:** The server-side logic must be implemented entirely in core PHP (without utilizing enterprise frameworks like Laravel or Symfony).
- **Database:** The system is constrained to a relational database architecture and must run on MySQL / MariaDB using the exact schema provided in `myhmsdb.sql`.

3.4.2 Execution Environment

- **Server Hosting:** The application is designed to be hosted on an Apache Web Server configured with PHP support. For local development and testing, it is constrained to the XAMPP environment.
- **PHP Version Dependency:** The server environment must support PHP 7.x (or compatible versions) to successfully execute the backend logic and SQL connection queries (`mysqli`).

3.4.3 Third-Party Library Dependencies

- **PDF Generation:** The automated generation of patient billing receipts is strictly dependent on the open-source **TCPDF** PHP library. Any modifications to the billing format are limited by the capabilities of this specific library.
- **Iconography:** The system relies on FontAwesome libraries (`font-awesome.min.css`) for UI iconography.

3.4.4 Client-Side Constraints

- **Browser Compatibility:** The application requires a modern web browser (e.g., Chrome, Firefox, Edge).

- **JavaScript Requirement:** The client's web browser must have JavaScript enabled. If JavaScript is disabled, critical functionalities such as input validation (e.g., alphanumeric name restrictions), Bootstrap UI toggles, and dynamic tab switching will fail.

3.4.5 Architectural Constraints

- **Monolithic Architecture:** The system is built as a monolithic web application rather than a microservices architecture. Client-side and server-side code are tightly coupled within the PHP files (e.g., HTML forms directly submitting to embedded PHP logic).
- **Session Management:** User state and authentication are managed entirely via native PHP sessions (\$_SESSION). Browsers must accept session cookies for the login modules to function securely.

3.5 Software System Attributes

This section specifies the attributes that are expected from the Hospital Management System and their implementations.

3.5.1 Accuracy

The accuracy requirements for the system are as follows:

Requirement ID	Description	Priority	Author
NFR001	The system shall ensure that all appointment and prescription data are stored and displayed accurately without data loss.	High	CHIN ZHEN HO

3.5.2 Security

The security requirements for the system are as follows:

Requirement ID	Description	Priority	Author
NFR002	The system shall restrict access to authorized users only by requiring login authentication.	High	CHIN ZHEN HO
NFR003	The system shall ensure that users can only access data according to their roles and permissions.	High	CHIN ZHEN HO
NFR004	The system shall terminate user sessions after logout.	High	CHIN ZHEN HO

3.5.3 Usability

The usability requirements for the system are as follows:

Requirement ID	Description	Priority	Author
NFR007	The system shall provide a user-friendly and easy-to-navigate interface.	Medium	CHIN ZHEN HO
NFR008	The system shall display clear validation	Medium	CHIN ZHEN HO

	messages for user input errors.		
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3.5.4 Availability

The availability requirements for the system are as follows:

Requirement ID	Description	Priority	Author
NFR009	The system shall be available 24/7 except during maintenance.	High	CHIN ZHEN HO